

Residual-wrench / parameter diagnostic

Linearized model: $\bar{w}(\theta + \delta) \approx \bar{w}(\theta) + J \delta$

Metric: same reference force/torque scaling as deterministic SG second-moment objective

Sample count: 658

Objective Jacobian rank: 14

Absorbable residual second-moment fraction: 1

Irreducible residual second-moment fraction: 0

Per-component absorbable fractions: [1.0, 1.0, 1.0, 1.0, 1.0, 1.0]

Remaining raw wrench mean [Fx,Fy,Fz,Mx,My,Mz]: [1.17554020e-15 -2.89493387e-16 3.88980496e-15
-3.41125999e-15

-1.28666288e-15 -2.11029932e-16]

Remaining raw wrench RMS [Fx,Fy,Fz,Mx,My,Mz]: [1.79259937e-15 4.10949857e-15 5.20168153e-15
3.45878148e-15

1.41939679e-15 2.59743672e-15]

Best local correction for the realized residual

log_mass_scale	-1
log_second_moment_cholesky_xx_scale	-0.5
log_second_moment_cholesky_yy_scale	-0.5
log_second_moment_cholesky_zz_scale	-0.5
normalized_second_moment_cholesky_yx_offset	-0.0003790761571
normalized_second_moment_cholesky_zx_offset	-0.0005385376537
normalized_second_moment_cholesky_zy_offset	+0.002814443777
cog_position_body_x_m	-5.317551795e-17
cog_position_body_y_m	-1.223036913e-16
cog_position_body_z_m	-7.976327693e-17
log_force_effectiveness_1	-1
log_force_effectiveness_2	-1
log_force_effectiveness_3	-1
log_force_effectiveness_4	-1

Mean-induced parameter bias under current iid residual model

log_mass_scale	bias=+0.02884361546	std=0.03065053262
log_second_moment_cholesky_xx_scale	bias=+0.09144475808	std=0.03622414047
log_second_moment_cholesky_yy_scale	bias=-0.05260376659	std=0.02677878711
log_second_moment_cholesky_zz_scale	bias=+6.426246846	std=3.407214154
normalized_second_moment_cholesky_yx_offset	bias=-0.003980486382	std=0.01329461063
normalized_second_moment_cholesky_zx_offset	bias=+0.1005534456	std=0.1453422114
normalized_second_moment_cholesky_zy_offset	bias=+0.09804863545	std=0.1638172281
cog_position_body_x_m	bias=+0.003913097807	std=0.002394051576
cog_position_body_y_m	bias=+0.004205401832	std=0.002035258445
cog_position_body_z_m	bias=-0.01257186685	std=0.003072967541
log_force_effectiveness_1	bias=-0.04434268005	std=0.03014442365
log_force_effectiveness_2	bias=-3.148069232e-05	std=0.03223789402
log_force_effectiveness_3	bias=+0.02015919582	std=0.0316723822
log_force_effectiveness_4	bias=-0.02008698423	std=0.03402498207

Physical parameter bias/std

mass_kg	bias=+0.06866237776	std=0.07296375353
inertia_xx_kg_m2	bias=+0.001394294221	std=0.002658398992
inertia_yy_kg_m2	bias=+0.01998130545	std=0.00650065091
inertia_zz_kg_m2	bias=+0.005050312532	std=0.005401945034
inertia_xy_kg_m2	bias=+0.0002525001244	std=0.0008585015766
inertia_xz_kg_m2	bias=-0.001908679284	std=0.002756307325
inertia_yz_kg_m2	bias=-0.001862410555	std=0.003099292715
cog_position_body_x_m	bias=+0.003913097807	std=0.002394051576

cog_position_body_y_m	bias=+0.004205401832	std=0.002035258445
cog_position_body_z_m	bias=-0.01257186685	std=0.003072967541
force_effectiveness_1	bias=-0.03973662443	std=0.02701319901
force_effectiveness_2	bias=-2.871559223e-05	std=0.02940628528
force_effectiveness_3	bias=+0.01782435479	std=0.02800408223
force_effectiveness_4	bias=-0.01780779952	std=0.03016431198

Parameter covariance in raw 14-D coordinate

```
[ 0.00093945515, 0.00042183058, -3.6643193e-05, 0.022454866, -7.4584919e-05, 0.00069912981,
 0.00030463745, -2.6754982e-06, 8.8852296e-06, -7.2057516e-06, 0.00086448008,
 0.00087283713, 0.0009093993, 0.00094740481]
[ 0.00042183058, 0.0013121884, -0.00013859126, 0.0047713295, 5.8116356e-05, 0.0011310341,
 0.0022364942, -2.3339703e-05, 3.4279289e-05, -6.0753549e-05, 0.00036863271,
 0.00019926067, 0.00042886491, 0.00059624866]
[-3.6643193e-05, -0.00013859126, 0.00071710344, -0.072098717, -1.7186168e-05, -0.00033097998,
 0.0016815886, 2.014578e-08, 1.0809753e-05, 4.5466933e-05, -7.4405464e-05,
 -0.00010940213, 1.7497746e-05, 3.8611257e-05]
[ 0.022454866, 0.0047713295, -0.072098717, 11.609108, -0.00055365098, -0.011179155,
 -0.32744397, 0.00074521185, -0.00264616, -0.0064237232, 0.027837828, 0.040015833,
 0.012978249, 0.0033791217]
[-7.4584919e-05, 5.8116356e-05, -1.7186168e-05, -0.00055365098, 0.00017674667,
 -0.0010244049, 1.3302538e-05, -8.383265e-06, -5.9202805e-07, -7.4156543e-06,
 -2.1555247e-05, -0.0001113158, -0.00010479415, -4.5709224e-05]
[ 0.00069912981, 0.0011310341, -0.00033097998, -0.011179155, -0.0010244049, 0.021124358,
 0.0040182595, 4.153379e-06, 5.896125e-05, -3.1752381e-05, 0.00038468951, 0.0005075498,
 0.00090785045, 0.00085654129]
[ 0.00030463745, 0.0022364942, 0.0016815886, -0.32744397, 1.3302538e-05, 0.0040182595,
 0.026836084, -0.00011121039, 0.0003153696, 9.5359886e-05, -0.00043845836, -0.0015510618,
 0.0010690046, 0.0020963093]
[-2.6754982e-06, -2.3339703e-05, 2.014578e-08, 0.00074521185, -8.383265e-06, 4.153379e-06,
 -0.00011121039, 5.7314829e-06, -1.3373688e-06, 9.7447985e-07, -1.966114e-05,
 2.6345228e-05, 1.5029055e-05, -2.8596996e-05]
[ 8.8852296e-06, 3.4279289e-05, 1.0809753e-05, -0.00264616, -5.9202805e-07, 5.896125e-05,
 0.0003153696, -1.3373688e-06, 4.1422769e-06, -2.2320026e-07, -1.3779805e-06,
 -1.4488507e-05, 1.9008698e-05, 3.068994e-05]
[-7.2057516e-06, -6.0753549e-05, 4.5466933e-05, -0.0064237232, -7.4156543e-06,
 -3.1752381e-05, 9.5359886e-05, 9.7447985e-07, -2.2320026e-07, 9.4431295e-06,
 -1.0871897e-05, -5.5747834e-06, -2.3672765e-06, -5.031969e-06]
[ 0.00086448008, 0.00036863271, -7.4405464e-05, 0.027837828, -2.1555247e-05, 0.00038468951,
 -0.00043845836, -1.966114e-05, -1.3779805e-06, -1.0871897e-05, 0.00090868628,
 0.00076872423, 0.00072742095, 0.00090262209]
[ 0.00087283713, 0.00019926067, -0.00010940213, 0.040015833, -0.0001113158, 0.0005075498,
 -0.0015510618, 2.6345228e-05, -1.4488507e-05, -5.5747834e-06, 0.00076872423,
 0.0010392818, 0.00087877441, 0.00067678981]
[ 0.0009093993, 0.00042886491, 1.7497746e-05, 0.012978249, -0.00010479415, 0.00090785045,
 0.0010690046, 1.5029055e-05, 1.9008698e-05, -2.3672765e-06, 0.00072742095,
 0.00087877441, 0.0010031398, 0.00089121629]
[ 0.00094740481, 0.00059624866, 3.8611257e-05, 0.0033791217, -4.5709224e-05,
 0.00085654129, 0.0020963093, -2.8596996e-05, 3.068994e-05, -5.031969e-06, 0.00090262209,
 0.00067678981, 0.00089121629, 0.0011576994]
```

Parameter second moment in raw 14-D coordinate

```
[ 0.0017714093, 0.003059428, -0.001553926, 0.20781106, -0.00018939654, 0.0035994547,
 0.0031327146, 0.00011019239, 0.00013018422, -0.00036982384, -0.00041452313,
 0.00087192911, 0.0014908634, 0.00036802356]
```

```
[ 0.003059428, 0.0096743321, -0.00494893, 0.59241792, -0.00030587826, 0.01032612,
0.011202528, 0.00033449258, 0.00041884124, -0.0012103849, -0.0036862729, 0.00019638192,
0.0022723177, -0.0012406008]
[-0.001553926, -0.00494893, 0.0034842597, -0.41014351, 0.00019220241, -0.00562047,
-0.0034761389, -0.00020582354, -0.00021041022, 0.00070679448, 0.0022581865,
-0.00010774613, -0.0010429519, 0.0010952623]
[ 0.20781106, 0.59241792, -0.41014351, 52.905757, -0.026133239, 0.63500211, 0.30264077,
0.025891744, 0.02437879, -0.087213643, -0.25711918, 0.03981353, 0.14252622, -0.1257048]
[-0.00018939654, -0.00030587826, 0.00019220241, -0.026133239, 0.00019259094, -0.0014246565,
-0.00037697872, -2.3959298e-05, -1.7331573e-05, 4.2626491e-05, 0.00015495019,
-0.00011119049, -0.00018503755, 3.4246743e-05]
[ 0.0035994547, 0.01032612, -0.00562047, 0.63500211, -0.0014246565, 0.031235354,
0.013877388, 0.00039762885, 0.00048182889, -0.0012958969, -0.0040741198, 0.00050438431,
0.0029349271, -0.0011632742]
[ 0.0031327146, 0.011202528, -0.0034761389, 0.30264077, -0.00037697872, 0.013877388,
0.036449619, 0.00027246351, 0.00072770351, -0.0011372945, -0.0047861976, -0.0015541485,
0.0030455862, 0.00012680795]
[ 0.00011019239, 0.00033449258, -0.00020582354, 0.025891744, -2.3959298e-05, 0.00039762885,
0.00027246351, 2.1043817e-05, 1.511878e-05, -4.8220465e-05, -0.00019317838,
2.6222041e-05, 9.391396e-05, -0.00010719933]
[ 0.00013018422, 0.00041884124, -0.00021041022, 0.02437879, -1.7331573e-05, 0.00048182889,
0.00072770351, 1.511878e-05, 2.1827682e-05, -5.3092952e-05, -0.00018785677,
-1.4620896e-05, 0.00010378622, -5.3783901e-05]
[-0.00036982384, -0.0012103849, 0.00070679448, -0.087213643, 4.2626491e-05, -0.0012958969,
-0.0011372945, -4.8220465e-05, -5.3092952e-05, 0.00016749497, 0.00054659837,
-5.1790123e-06, -0.000255806, 0.00024749892]
[-0.00041452313, -0.0036862729, 0.0022581865, -0.25711918, 0.00015495019, -0.0040741198,
-0.0047861976, -0.00019317838, -0.00018785677, 0.00054659837, 0.0028749596,
0.00077012017, -0.00016649182, 0.0017933328]
[ 0.00087192911, 0.00019638192, -0.00010774613, 0.03981353, -0.00011119049, 0.00050438431,
-0.0015541485, 2.6222041e-05, -1.4620896e-05, -5.1790123e-06, 0.00077012017,
0.0010392828, 0.00087813979, 0.00067742216]
[ 0.0014908634, 0.0022723177, -0.0010429519, 0.14252622, -0.00018503755, 0.0029349271,
0.0030455862, 9.391396e-05, 0.00010378622, -0.000255806, -0.00016649182, 0.00087813979,
0.001409533, 0.00048627884]
[ 0.00036802356, -0.0012406008, 0.0010952623, -0.1257048, 3.4246743e-05, -0.0011632742,
0.00012680795, -0.00010719933, -5.3783901e-05, 0.00024749892, 0.0017933328,
0.00067742216, 0.00048627884, 0.0015611863]
```

Physical parameter covariance

```
[ 0.0053237093, 5.6672752e-05, 0.00019769564, 0.00011854245, 1.1413274e-05,
-3.1580567e-05, -1.375288e-05, -6.3690376e-06, 2.1151336e-05, -1.715333e-05, 0.0018441369,
0.0018952908, 0.0019140989, 0.0019994026]
[ 5.6672752e-05, 7.0670852e-06, 5.4646361e-06, -1.4703813e-06, 1.8849111e-07,
1.0790726e-06, 3.8131645e-06, 9.559351e-07, -1.9915371e-06, -2.3217167e-06,
2.3154231e-05, 3.3636849e-05, 1.6516377e-05, 8.1001647e-06]
[ 0.00019769564, 5.4646361e-06, 4.2258462e-05, 8.5784135e-06, -4.4722968e-07,
-2.5061933e-06, 2.4227708e-06, -2.0623473e-06, 1.0481296e-06, -1.6022748e-05,
7.4429233e-05, 6.9977212e-05, 6.3543342e-05, 7.2017669e-05]
[ 0.00011854245, -1.4703813e-06, 8.5784135e-06, 2.918101e-05, -3.5070753e-07,
-1.9656975e-06, -9.5578344e-06, -3.0147382e-06, 5.8210998e-06, -2.0030001e-06,
3.4114088e-05, 1.0641969e-05, 5.0989856e-05, 7.2717642e-05]
[ 1.1413274e-05, 1.8849111e-07, -4.4722968e-07, -3.5070753e-07, 7.3702496e-07,
-1.2529913e-06, 1.7725853e-08, 5.4235697e-07, 3.654288e-08, 4.817203e-07,
1.2308654e-06, 6.5463139e-06, 5.9632686e-06, 2.5902357e-06]
```

[-3.1580567e-05, 1.0790726e-06, -2.5061933e-06, -1.9656975e-06, -1.2529913e-06,
7.5972301e-06, 1.4488573e-06, -7.8284502e-08, -1.118792e-06, 6.033665e-07, -6.5439233e-06,
-8.7830748e-06, -1.5229536e-05, -1.4410536e-05]
[-1.375288e-05, 3.8131645e-06, 2.4227708e-06, -9.5578344e-06, 1.7725853e-08,
1.4488573e-06, 9.6056153e-06, 2.104627e-06, -5.967647e-06, -1.7991126e-06, 7.4237302e-06,
2.6758527e-05, -1.7894538e-05, -3.5175208e-05]
[-6.3690376e-06, 9.559351e-07, -2.0623473e-06, -3.0147382e-06, 5.4235697e-07,
-7.8284502e-08, 2.104627e-06, 5.7314829e-06, -1.3373688e-06, 9.7447985e-07,
-1.7618857e-05, 2.4031201e-05, 1.3288388e-05, -2.5352217e-05]
[2.1151336e-05, -1.9915371e-06, 1.0481296e-06, 5.8210998e-06, 3.654288e-08, -1.118792e-06,
-5.967647e-06, -1.3373688e-06, 4.1422769e-06, -2.2320026e-07, -1.234844e-06,
-1.3215912e-05, 1.6807108e-05, 2.7207683e-05]
[-1.715333e-05, -2.3217167e-06, -1.6022748e-05, -2.0030001e-06, 4.817203e-07, 6.033665e-07,
-1.7991126e-06, 9.7447985e-07, -2.2320026e-07, 9.4431295e-06, -9.7425882e-06,
-5.0851234e-06, -2.0930982e-06, -4.4610129e-06]
[0.0018441369, 2.3154231e-05, 7.4429233e-05, 3.4114088e-05, 1.2308654e-06,
-6.5439233e-06, 7.4237302e-06, -1.7618857e-05, -1.234844e-06, -9.7425882e-06,
0.00072971292, 0.00062836662, 0.00057636215, 0.00071708479]
[0.0018952908, 3.3636849e-05, 6.9977212e-05, 1.0641969e-05, 6.5463139e-06,
-8.7830748e-06, 2.6758527e-05, 2.4031201e-05, -1.3215912e-05, -5.0851234e-06,
0.00062836662, 0.00086472961, 0.00070874747, 0.0005472967]
[0.0019140989, 1.6516377e-05, 6.3543342e-05, 5.0989856e-05, 5.9632686e-06,
-1.5229536e-05, -1.7894538e-05, 1.3288388e-05, 1.6807108e-05, -2.0930982e-06,
0.00057636215, 0.00070874747, 0.00078422862, 0.00069858499]
[0.0019994026, 8.1001647e-06, 7.2017669e-05, 7.2717642e-05, 2.5902357e-06,
-1.4410536e-05, -3.5175208e-05, -2.5352217e-05, 2.7207683e-05, -4.4610129e-06,
0.00071708479, 0.0005472967, 0.00069858499, 0.00090988572]

Physical parameter second moment

[0.010038231, 0.00015240831, 0.0015696596, 0.00046530892, 2.8750533e-05, -0.00016263503,
-0.00014163042, 0.00026231356, 0.00030990422, -0.0008803676, -0.00088427418,
0.0018933191, 0.0031379615, 0.00077667671]
[0.00015240831, 9.0111416e-06, 3.3324455e-05, 5.5712403e-06, 5.4055058e-07,
-1.5821879e-06, 1.2164162e-06, 6.4119448e-06, 3.8720303e-06, -1.9850598e-05,
-3.2250315e-05, 3.3596811e-05, 4.1368772e-05, -1.6729147e-05]
[0.0015696596, 3.3324455e-05, 0.00044151103, 0.00010949025, 4.5980524e-06,
-4.0644097e-05, -3.4790623e-05, 7.6126455e-05, 8.5077548e-05, -0.00026722506,
-0.0007195604, 6.9403437e-05, 0.00041969722, -0.00028380541]
[0.00046530892, 5.5712403e-06, 0.00010949025, 5.4686667e-05, 9.2449701e-07,
-1.1605124e-05, -1.896359e-05, 1.6747629e-05, 2.7059693e-05, -6.5494857e-05,
-0.00016656828, 1.0496946e-05, 0.00014100842, -1.7217312e-05]
[2.8750533e-05, 5.4055058e-07, 4.5980524e-06, 9.2449701e-07, 8.0078127e-07,
-1.7349331e-06, -4.5253304e-07, 1.5304147e-06, 1.0984074e-06, -2.6926776e-06,
-8.8026372e-06, 6.5390632e-06, 1.046392e-05, -1.9062359e-06]
[-0.00016263503, -1.5821879e-06, -4.0644097e-05, -1.1605124e-05, -1.7349331e-06,
1.1240287e-05, 5.0036017e-06, -7.5471332e-06, -9.1455554e-06, 2.4599028e-05,
6.9300549e-05, -8.728266e-06, -4.9250512e-05, 1.9578842e-05]
[-0.00014163042, 1.2164162e-06, -3.4790623e-05, -1.896359e-05, -4.5253304e-07,
5.0036017e-06, 1.3074188e-05, -5.1831676e-06, -1.3799832e-05, 2.1614865e-05,
8.1429639e-05, 2.6812007e-05, -5.1090805e-05, -2.0097742e-06]
[0.00026231356, 6.4119448e-06, 7.6126455e-05, 1.6747629e-05, 1.5304147e-06,
-7.5471332e-06, -5.1831676e-06, 2.1043817e-05, 1.511878e-05, -4.8220465e-05,
-0.00017311215, 2.3918834e-05, 8.3036832e-05, -9.5035878e-05]
[0.00030990422, 3.8720303e-06, 8.5077548e-05, 2.7059693e-05, 1.0984074e-06,
-9.1455554e-06, -1.3799832e-05, 1.511878e-05, 2.1827682e-05, -5.3092952e-05,

-0.00016834332, -1.3336673e-05, 9.1765682e-05, -4.768127e-05]
[-0.0008803676, -1.9850598e-05, -0.00026722506, -6.5494857e-05, -2.6926776e-06,
2.4599028e-05, 2.1614865e-05, -4.8220465e-05, -5.3092952e-05, 0.00016749497,
0.00048982096, -4.7241148e-06, -0.00022617851, 0.00021941627]
[-0.00088427418, -3.2250315e-05, -0.0007195604, -0.00016656828, -8.8026372e-06,
6.9300549e-05, 8.1429639e-05, -0.00017311215, -0.00016834332, 0.00048982096,
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[0.0018933191, 3.3596811e-05, 6.9403437e-05, 1.0496946e-05, 6.5390632e-06, -8.728266e-06,
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